

Ann Rose Joju Edakkalathur

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Summary

Backend Developer with hands-on experience designing, building, and optimizing scalable server-side applications using Java, Spring Boot, REST APIs, and cloud technologies. Skilled in database design, API integration, microservices, and performance tuning, with a strong foundation in data structures, algorithms, and secure backend architecture. Proven ability to improve system efficiency, enhance application reliability, and collaborate with cross-functional teams to deliver high-quality, production-ready solutions.

Education

Mercy University, Dobbs Ferry, New York

Sep 2024-Dec 2025

Master's in computer science

GPA 3.9

APJ Abdul Kalam technological University, Kerala, India

Jun 2018-Jun 2022

Bachelor of Technology in Electronics and Communication Engineering

GPA 3.2

Experience

Centre for STEM Education, Mercy University, Tarrytown, NY

Sep 2025-Dec 2025

Data Analyst & STEM Program Assistant

- Support STEM outreach initiatives including Westchester STEM Ambassadors, USA Blue Schools, Saturday STEM Academy, Wednesday STEM Academy, and STEM Teacher Training
- Analyzed student engagement, attendance, and program performance data to support Verizon STEM initiatives and inform program improvements.
- Coordinating logistics, managing tech resources, handling office tasks, and collaborating with faculty to deliver inclusive, hands-on learning experiences for K–12 students.

Mercy University, Dobbs Ferry, NY

Jan 2025- Dec 2025

Teaching Assistant

- Assisted students in understanding fundamental programming concepts, algorithm design, and problem-solving using Python
- Provided grading support, constructive feedback, and mentorship to improve student comprehension and coding proficiency.

INFOSYS LIMITED, India

Nov 2022- Sep 2024

System Engineer

- Designed and deployed Java Spring Boot features and REST APIs for an internal employee attendance portal using Git and Maven, improving operational efficiency and user experience by 25%.
 - Led Java version upgrade and implemented new features for Medicare software at Beacon Healthcare Systems (USA), resulting in a 30% improvement in accuracy and enhanced system performance through rigorous testing and end-to-end development.
 - Developed secure backend modules using Java, Spring Boot, and Spring Security for Neom's social media platform, achieving a 60% reduction in response times and improved application security.
 - Delivered custom web and mobile solutions for The Line (Neom, Saudi Arabia) using Microsoft Power Platform, including website creation, workflow automation, and mobile app development all with zero downtime during implementation.
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Academic Projects

Personalized News Recommendation System – MIND Dataset

Sep 2025- Dec2025

- Built and compared popularity-based, content-based (TF-IDF), collaborative filtering (SVD), and hybrid deep learning models using user behaviour and article metadata from the MIND-small dataset
- Pre-processed and merged news and behaviour logs, engineered features, and evaluated models using Precision@K, MAP, NDCG, and AUC to identify optimal personalization strategies

Drive Smart - Enhancing Vehicle Safety and Efficiency with Predictive Maintenance

Jan 2025- May 2025

- Developed a machine learning model to predict vehicle maintenance needs and evaluated multiple algorithms.
- Deployed the model as a Flask-based web app for real-time predictions, enabling proactive maintenance scheduling to improve cost efficiency & fleet safety

Cancer Detection Using AI -Machine learning to analyse MRI brain scans and predict tumour

Sep 2021-May 2022

- Pre-processed MRI brain scan datasets and implemented ML models for tumor classification
 - Achieved 80% accuracy in predicting tumor presence in new images
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Technical Skills

Certifications: Microsoft certified: Azure Fundamentals- AZ 900, Microsoft certified: Azure Administrator Associate-AZ 104,

Microsoft certified: Azure AI Fundamentals-AI 900, Infosys certified Database and SQL Professional

Programming Languages: Java, Python, HTML, SQL, CSS, R, JavaScript

Operating System: Microsoft Windows, Linux, Android, iOS

Software: GitHub, Postman, IntelliJ, VS Code, Android Studio, Eclipse, Tableau, MATLAB, Google Workspace, Microsoft Power Automate, Power Apps, Microsoft Office Suite, MS applications, Spring Boot

Cloud Platforms: Microsoft Azure (Databricks, ML Studio, Compute, Storage), AWS (EC2, S3, Lambda), Google Cloud Platform (GCP)

Machine Learning & AI: Natural Language Processing (NLP), Predictive Modelling, Statistical Analysis, Data Preprocessing, Feature Engineering, Logistic Regression, Decision Trees, Random Forest, SVM, XGBoost, TensorFlow, PyTorch, spaCy, NLTK, LangChain, Spark, Kafka

Model Evaluation: Confusion Matrix, ROC Curve, AUC, Cross-Validation, Hyperparameter Tuning

Data Tools: Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Jupyter Notebook, Google Colab, PyCharm